



Indian Association for the Cultivation of Science
(Deemed to be university under the *de novo* category)

Integrated BSMS Program

End-Semester Examination-2025 (Autumn Semester 3-BSMS)

Subject: Quantum and Thermal Physics
Full Marks: 50

Subject Code(s): PHS 2101
Time Allotted: 3hr

Answer any five questions from the seven.

1. (a) The distribution of molecular speeds of an ideal gas is given by

$$F(v)dv = 4\pi C \exp\left[-(1/2)mv^2/kT\right]dv$$

Determine the constant C . Sketch approximately the shapes of this distribution function $F(v)$ vs v for temperatures $T_1 > T_2 > T_3$ on the same axes [2+3].

- (b) Calculate the most probable speed of a molecule of nitrogen gas (molecular weight 28) at room temperature ($T \approx 300$) degrees kelvin, given that Avogadro number is 6×10^{23} . Treat nitrogen as an ideal gas.[5]

2. (a) In the Meyer-Gerlach experiment on the photoelectric effect, neutral dust is irradiated with ultraviolet light to release electrons from the dust particles, producing an electric current. An appropriate electric field is then applied to balance the effect of gravity on the particles, so they float in mid-air. Keeping this electric field the same, what would one observe, if the system is once again irradiated with UV light? Precisely what is the quantity measured in this experiment, which confirms that UV light is composed of photons rather than classical EM waves? [2+3]

- (b) In the Feynman electron double slit experiment, with both slits open, an interference pattern is observed on the screen. If a claim is made that the interference appears due to multiple electrons interacting with each other, what must you do to ensure that this is not the case? Do you see the same pattern on the screen, if a photon source is placed between the slits, and suitable photodetectors are placed close to the slits, to ascertain which slit the electron went through? Why? Explain your answer as quantitatively as you can. [1+1+3]

3. (a) Consider a one dimensional simple harmonic oscillator of mass m and frequency ω . Using classical mechanics, what does one get for the *minimum* energy of this system ? How does this answer change if we go over to quantum mechanics? Appealing to one of the principle tenets of quantum mechanics, argue why this change occurs. [1+1+3]
- (b) In any discrete basis on the state space in quantum mechanics, probability amplitudes are written as $\langle i|j\rangle$, with $i, j = 1, 2, \dots, N$. How does this probability amplitude change if we make simultaneous linear transformations on both the state ket vector $|j\rangle$ and the bra state vector $\langle i|$? How does this probability amplitude change, if we make the same *unitary* transformation on both the ket and bra vectors ? [3+2]
4. (a) What is the key assumption which leads to the time-evolution equation for a state ket vector which depends on time $|\psi(t)\rangle$, namely

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

Deduce from the above equation the TDSE for the wave-function $\psi_I(t)$, $I = 1, 2$ in a discrete two dimensional basis. [2+3]

- (b) Consider the transition between the two states of the ammonia molecule, in which the nitrogen atom is above ($|C'\rangle$) or below ($|T'\rangle$) the plane of the three hydrogen atoms, keeping in mind the 2×2 Hamiltonian appropriate for such transitions. If we begin with an ammonia gas in which all the molecules are in the $|T'\rangle$ state at $t = 0$, how much time must elapse after which we shall find that the gas has 50% of those molecules remaining in the $|T'\rangle$ state ? [5]

5. (a) What is the result of $\hat{p}\psi(p)$ where $\psi(p) = \langle p|\psi\rangle$? Make the guess that the action of $\hat{x}\psi(p) = i\hbar(\partial/\partial p)\psi(p)$. How would you verify if your guess is correct ? [1+4]
- (b) If an electron, with spin $1/2$ is moving very slowly in a constant uniform magnetic field $\vec{B}$, it is described in the spin $1/2$ basis by a 2×2 Hamiltonian matrix given by $\underline{H} = (q_e \hbar / 2m) \underline{\vec{\sigma}} \cdot \vec{B}$, where $\underline{\vec{\sigma}}$ are the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

How would you choose $\vec{B}$ so that the solution of the TDSE in the spin $(1/2)$ basis gives you *stationary* states ? What are the energy eigenvalues of the Hamiltonian in this case ? [3+2]

6. (a) Using the theorem that *any* 2×2 matrix can be written as a complex linear combination of the 2×2 identity matrix and the Pauli matrices given above, write down the Hamiltonian for the ammonia molecule which you have used in Q4 as such a linear combination which represents the Hamiltonian for a spin $(1/2)$ electron in a magnetic field, for an appropriate choice of the magnetic field. [5]

- (b) Show explicitly that $\sigma_x \sigma_y + \sigma_y \sigma_x = 0$, $\sigma_x \sigma_y - \sigma_y \sigma_x = 2i\sigma_z$. [5]

7. (a) Is the wave function for a free particle solution of the time-independent Schrödinger equation, normalizable ? Why ? A Gaussian wave packet is described by a smearing function $f(q)$ which is identical to the distribution function $F(q)$ given in Q1. Show that the wave packet wave function $\Psi(x) = \int (dq/2\pi) \exp[iqx] f(q)$ is normalizable. [1+4]

- (b) For a particle in a potential barrier $V = 0, x < 0; V = V_0, x > 0$ show that for a free particle approaching the barrier from the $x < 0$ side of the x -axis, the reflection coefficient $\mathcal{R} = 1$. Is there any transmission of the incident wave function in the $x > 0$ region ?
[4+1]